

1. Complete the truth table for $F(A, B, C) = A'B + B'C' + AC$.
2. Write $F(A, B, C) = \Sigma m(0, 2, 5, 7)$ in expanded sum-of-minterms notation.
3. Write $F(A, B, C) = \Pi M(1, 3, 4, 6)$ in expanded product-of-maxterms notation.
4. Find the complement of $F = x'y + z(w + x)$.
5. Simplify $F = A'BC + A'BC' + AB'C' + ABC'$ to a minimum number of literals.
6. Simplify $G = (x + z)(x + z')(x' + y)$ to a minimum number of literals.
7. Simplify $H = w'x'y + w'xy + wx'y + wxy$ to a minimum number of literals.
8. Determine whether $pq + p'r + qr = pq + p'r$ is true using Boolean algebra.
9. Compute the bitwise NOR of $P = 01011100$ and $Q = 11001010$.
10. Draw a gate-level logic diagram for $Y = (A' + B)(C + D') + AC$ using only NOT, AND, and OR gates.